

# Structures at six: the Eisenstein geometry of $E_6$ , a parity obstruction for bilinear $\text{AME}(4, d)$ constructions, and the anatomy of two $\text{AME}(4, 6)$ representatives

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## Abstract

Local dimension six carries two famous exceptions: there is no pair of orthogonal Latin squares of order six (Tarry), and the maximal number of mutually unbiased bases in  $\mathbb{C}^6$  remains unknown. The first exception was resolved quantum-mechanically in 2022 by the construction of an absolutely maximally entangled state  $\text{AME}(4, 6)$ , equivalently a 2-unitary matrix of order 36. This note collects four independent results, all verified computationally from first principles.

(i) We give an explicit realization of the  $\mathbb{Z}[\omega]$ -module structure of the  $E_6$  root lattice through the fourth power of a Coxeter element, and verify that the resulting 12 complex lines in  $\mathbb{C}^3$  decompose into exactly 4 orthogonal frames that are pairwise mutually unbiased. Thus  $E_6$  realizes the *complete* MUB system of a qutrit, and the count of positive roots factors as  $36 = 12 \times 3$ .

(ii) We prove a parity obstruction: a flat 2-unitary of order  $d^2$  whose phase is a bilinear form over  $\mathbb{Z}/d$  exists *if and only if*  $d$  is odd. The proof is two lines and uses only that the units of  $\mathbb{Z}/d$  are odd for even  $d$ . This explains why the finite-field construction available at every odd  $d$  has no analogue at  $d = 6$ , and shows the obstruction is one of parity rather than of non-prime-power order.

(iii) We reconstruct and independently verify two Butson-type  $\text{AME}(4, 6)$  representatives with all phases in  $\mu_6 = \mathbb{Z}[\omega]^\times$ , and show that their phase function is not a polynomial of degree  $\leq 2$  over  $\mathbb{Z}/6$ , is not translation covariant, and does not factor through the Chinese Remainder decomposition  $\mathbb{Z}/6 \cong \mathbb{Z}/2 \times \mathbb{Z}/3$ .

(iv) We compute the automorphism group of the golden  $\text{AME}(4, 6)$  representative and find it to be  $\mathbb{Z}/2$  at the level of the modulus pattern, with the nontrivial element *failing* to lift to a monomial symmetry of the state. In particular the natural conjecture that the golden state is covariant for the exceptional action of  $\text{PGL}_2(\mathbb{F}_5)$  on the six points of  $\mathbb{P}^1(\mathbb{F}_5)$  is false for this representative. We show instead that  $\varphi$  enters *locally*, through the  $2 \times 2$  block relations, as the icosahedral vertex direction  $(1, \varphi)$ .

(v) We record a companion arithmetic thread attached to the same modulus: a proof via the valence formula that the modular-zero polynomial in weight  $12m$  has degree  $m(m-1)$ ; an independent computation of the weight-36 Hecke cubic, whose discriminant shares the 12-digit prime 269461929553 with the sextic field of modular zeros; and unconditional numerics for  $L(s, \chi_{-3})$ , the  $L$ -function of the character underlying the Eisenstein master identity.

## 1 Introduction

Two exceptional phenomena meet at local dimension six.

The first is combinatorial. Euler conjectured in 1779, and Tarry proved in 1900, that no two orthogonal Latin squares of order six exist; 6 is the unique order  $> 2$  for which this fails [1, 2, 3]. The best classical configuration attains 34 of the 36 officer pairs. In 2022 Rather et al. [4] showed that the problem admits a quantum solution: an absolutely maximally entangled state of four

six-level systems,  $\text{AME}(4, 6)$ , equivalently a 2-unitary matrix of order 36, equivalently a perfect tensor with four legs of dimension six. The amplitudes of their solution involve the golden ratio  $\varphi = (1 + \sqrt{5})/2$  and phases that are 20th roots of unity; it is known as the *golden* AME state. Subsequently it was shown that infinitely many inequivalent solutions exist [6], and two further constructions produced representatives that are Butson-type complex Hadamard matrices with all 1296 entries of equal modulus and phases in  $\mu_6$  [7, 8].

The second exception is measurement-theoretic. In prime-power dimension  $d$  there exist  $d + 1$  mutually unbiased bases (MUBs), and this is maximal. In dimension six only three MUBs are known, and whether four exist is a long-standing open problem. Six is the smallest dimension where the question is open.

That the same dimension hosts both exceptions is folklore-level motivation for asking whether the objects that repair the first say anything about the second. This note is a report of what we could establish by direct computation, kept scrupulously separate from what we could not. Section 3 treats an Eisenstein structure in dimension three that turns out to be exactly a complete MUB system; Section 4 isolates the arithmetic obstruction at six and shows it is a parity phenomenon valid for all even  $d$ ; Section 5 dissects two  $\text{AME}(4, 6)$  representatives. Appendix A tabulates the epistemic status of every claim.

All computations were performed independently of the cited literature: root systems were generated from Cartan matrices by breadth-first reflection closure, and the AME matrices were rebuilt from their defining data and re-verified, rather than assumed.

## 2 Preliminaries

Let  $d \geq 2$  and write  $N = d^2$ . Index  $\mathbb{C}^N \otimes \mathbb{C}^N$  by four indices, and for  $X \in \mathbb{C}^{N \times N}$  define the *reshuffling* and *partial transpose*

$$X_{jk;lm}^R = X_{jl;km}, \quad X_{jk;lm}^\Gamma = X_{jm;lk}.$$

**Definition 1.**  $U \in \mathbb{U}(d^2)$  is *2-unitary* if  $U$ ,  $U^R$  and  $U^\Gamma$  are all unitary. Equivalently the associated four-index tensor  $\psi_{jklm} = d^{-1}U_{jk;lm}$  is a *perfect tensor*: every reduction to two of the four legs is maximally mixed,  $\text{tr} \rho^2 = d^{-2}$  for all six bipartitions. Equivalently  $|\psi\rangle \in \text{AME}(4, d)$  [10].

**Definition 2.**  $H$  of order  $N$  is a *complex Hadamard matrix* if  $HH^\dagger = N\mathbb{I}$  and  $|H_{jk}| = 1$  for all  $j, k$ ; it is of *Butson type*  $\text{BH}(N, q)$  if in addition every entry is a  $q$ -th root of unity [12].

We call a 2-unitary *flat* if all  $d^2 \times d^2$  entries have modulus  $d^{-1}$ , i.e. if  $dU$  is a complex Hadamard matrix. It is known that 2-unitaries do not exist for  $d = 2$  [11]; for  $d \neq 2, 6$  they are furnished by orthogonal Latin squares as permutation matrices, and  $d = 6$  is the case resolved in [4].

Throughout,  $\omega = e^{2\pi i/3}$ ,  $\mathbb{Z}[\omega]$  is the ring of Eisenstein integers,  $\mu_k$  is the group of  $k$ -th roots of unity, and  $\mathbb{Z}[\omega]^\times = \mu_6$ .

## 3 The Eisenstein geometry of $E_6$ and the qutrit MUB system

The number 36 occurs twice in our story: as  $|\Phi^+(E_6)|$  and as  $6^2$ . The following makes the first occurrence structurally transparent. That the  $E_6$  root system carries a  $\mathbb{Z}[\omega]$ -structure related to the Hessian configuration of 12 lines is classical, arising from the theory of complex reflection groups; that the Hesse configuration encodes the qutrit MUB system is known in quantum information. Our contribution is an explicit, uniform realization through the Coxeter element together with complete verification of the resulting geometry.

**Theorem 3.** *Let  $\Phi$  be the root system of  $E_6$  with  $W = W(E_6)$ , Coxeter element  $c$ , and invariant bilinear form  $B$  normalized so that  $B(r, r) = 2$  for roots. Put  $g = c^4$ . Then:*

(a)  $g^3 = 1$ ,  $g$  preserves  $B$ , and  $g$  is fixed-point-free:  $\det(g - 1) = 27 \neq 0$ .

(b)  $J := (2g + 1)/\sqrt{3}$  satisfies  $J^2 = -1$  and  $B(Ju, v) = -B(u, Jv)$ , so  $h(u, v) := B(u, v) + iB(u, Jv)$  is a Hermitian form making the ambient space  $\mathbb{C}^3$  and the  $E_6$  lattice a rank-3  $\mathbb{Z}[\omega]$ -module, with  $h(r, r) = 2$  for roots.

(c) The group  $\mu_6 = \{\pm 1, \pm g, \pm g^2\}$  acts freely on the 72 roots, with exactly 12 orbits of size 6. Hence

$$72 = 12 \times 6 = \#\{\omega\text{-lines}\} \times |\mathbb{Z}[\omega]^\times|, \quad 36 = |\Phi^+| = 12 \times 3.$$

(d) Let  $L_1, \dots, L_{12} \subset \mathbb{C}^3$  be the resulting complex lines. Then for  $i \neq j$  the quantity  $|h(r_i, r_j)|^2$  takes exactly two values, 0 and  $4/3$ . Each line is orthogonal to exactly two others, so the 12 lines partition into 4 orthogonal frames of 3; and for lines in different frames the normalized overlap is

$$\frac{|h(r_i, r_j)|^2}{h(r_i, r_i)h(r_j, r_j)} = \frac{4/3}{4} = \frac{1}{3}.$$

Consequently the 12 lines form  $d + 1 = 4$  pairwise mutually unbiased bases of  $\mathbb{C}^3$ : the complete MUB system of a qutrit.

*Proof (computational).* The 72 roots were generated from the  $E_6$  Cartan matrix by closing the six simple roots under simple reflections; all have  $B$ -norm 2. The Coxeter element has order 12, consistent with  $h(E_6) = 12$ ; since the exponents of  $E_6$  are 1, 4, 5, 7, 8, 11, none is divisible by 3, so  $c^4$  has no eigenvalue 1 and is fixed-point-free, which is what (a) records numerically as  $\det(g - 1) = 27$ . Items (b)–(d) were verified entrywise; the orthogonality graph on the 12 lines is 3-regular with components  $\{0, 7, 11\}$ ,  $\{1, 6, 10\}$ ,  $\{2, 5, 8\}$ ,  $\{3, 4, 9\}$  in the ordering produced by our enumeration, and all  $\binom{12}{2} - 4\binom{3}{2} = 66 - 12 = 54$  cross-frame pairs give exactly  $1/3$ .  $\square$

*Remark 4.* The identity  $g^2 + g + 1 = 0$  on the ambient space is what allows  $J = (2g + 1)/\sqrt{3}$  to square to  $-1$ ; the factor  $\sqrt{3}$  is the different of  $\mathbb{Q}(\omega)$ , and  $\det(g - 1) = 27 = 3^3$  is the norm of  $(1 - \omega)^3$  up to units. The chain  $A_1$  ( $h=2$ ),  $A_2$  ( $h=3$ ),  $G_2$  ( $h=6$ ),  $F_4$  ( $h=12$ ),  $E_8$  ( $h=30$ ) of root systems whose exponents are exactly the totatives of  $h$  was also verified; it places the mod-6 unit pattern  $\{1, 5\}$  as the exponent set of  $G_2$ .

*Remark 5* (The two 36s are not the same 36). Since  $|\Phi^+(E_6)| = 36 = 6^2$ , it is natural to ask whether the root system can index a 2-unitary of order 36 and so produce AME(4, 6) representatives. It cannot, for structural reasons we record here so that the coincidence is not pursued further.

First, the index set has the wrong factorization. The  $\mu_6$ -action of Theorem 3 meets the positive roots in exactly 3 roots per  $\omega$ -line, so  $36 = 12 \times 3$ ; and relative to a marking the 36 positive roots — equivalently the 36 nonsingular points of the  $O_6^-(2)$  quadric, or the 36 double-sixes on a cubic surface — fall into types  $1 + 15 + 20$ . Neither is a  $6 \times 6$  product. Since the reshuffling and partial-transpose conditions are defined only *after* an identification of the index set with  $\{1, \dots, 6\}^2$ , and  $E_6$  supplies none canonically, 2-unitarity is not even a well-posed question for a root-indexed table: we find  $\|U^R(U^R)^\dagger - \mathbb{I}\|_F$  ranging over 5.75–6.50 across random identifications of the same matrix.

Second, the naive tables fail ordinary unitarity, which *is* identification-independent. For  $U_{rs} = \frac{1}{6}e^{2\pi i f B(r, s)}$  the best case is  $f = \frac{1}{2}$ , where  $B(r, s) \in \{-1, 0, 1, 2\}$  makes the table real and  $\pm 1$ -valued; its Gram matrix has entries  $\{-\frac{1}{9}, \frac{1}{9}, 1\}$  and spectrum  $\frac{4}{9}$  (multiplicity 21),  $\frac{16}{9}$  (multiplicity 15). A two-point spectrum, neither point equal to 1: this is a regular two-graph, provably not unitary and not perturbatively close to one.

Third, the Hermitian refinement is ill-posed. Using  $\arg h(r, s)$  as a phase requires  $h(r, s) \neq 0$ , but  $h$  vanishes on exactly 216 of the 1296 pairs — uniformly 6 per row — so no canonical flat table exists. The variant using  $B(r, Js)$  fares no better: because  $J$  carries a factor  $1/\sqrt{3}$ , one finds  $B(r, Js) \in \{0, \pm \frac{1}{\sqrt{3}}, \pm \frac{2}{\sqrt{3}}, \pm \sqrt{3}\}$ , never an integer, so  $e^{2\pi i f B(r, Js)}$  is not a root of unity and no

Butson table results. Scoring the well-defined variants gives  $\|UU^\dagger - \mathbb{I}\|_F$  of 16.64, 15.89 and 13.44, all worse than the 3.94 of the real  $\pm 1$  table.

The positive reading is Theorem 3 itself: the quantum object  $E_6$  carries is a *qutrit* system — 12 lines in  $\mathbb{C}^3$  forming 4 mutually unbiased bases — and 36 counts  $12 \times 3$  there. Dimension six is not what this 36 is about.

## 4 A parity obstruction for bilinear 2-unitaries

At every odd  $d$  there is a one-line construction of a perfect tensor,  $|\psi\rangle \propto \sum_{i,j} |i, j, i+j, i+2j\rangle$  over  $\mathbb{Z}/d$ . Its Fourier dual is a flat 2-unitary whose phase is a bilinear form. The following theorem determines exactly when such an object exists, and shows the failure at  $d = 6$  is a parity phenomenon.

For  $M = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix} \in M_2(\mathbb{Z}/d)$  and  $\zeta = e^{2\pi i/d}$ , define  $U^{(M)} \in \mathbb{C}^{d^2 \times d^2}$  by

$$U_{(x,y),(u,v)}^{(M)} = \frac{1}{d} \zeta^{x(m_{11}u + m_{12}v) + y(m_{21}u + m_{22}v)}. \quad (1)$$

**Lemma 6.**  $U^{(M)}$  is 2-unitary if and only if all three of

$$\det M = m_{11}m_{22} - m_{12}m_{21}, \quad m_{11}m_{22}, \quad m_{12}m_{21}$$

are units of  $\mathbb{Z}/d$ .

*Proof.* Write the exponent as  $E = m_{11}xu + m_{12}xv + m_{21}yu + m_{22}yv$ .

*Unitarity of  $U$ .* For row indices  $(x, y), (x', y')$  with  $(s, t) = (x - x', y - y')$ ,

$$(UU^\dagger)_{(x,y),(x',y')} = \frac{1}{d^2} \sum_{u,v} \zeta^{(sm_{11}+tm_{21})u} \zeta^{(sm_{12}+tm_{22})v},$$

which by orthogonality of characters vanishes unless  $(s, t)M \equiv 0$ , in which case it equals 1. Hence  $U$  is unitary iff  $(s, t)M \equiv 0 \Rightarrow (s, t) = 0$ , i.e. iff  $\det M \in (\mathbb{Z}/d)^\times$ .

*Unitarity of  $U^R$ .* Reading  $E$  with row index  $(x, u)$  and column index  $(y, v)$ , the terms  $m_{11}xu$  and  $m_{22}yv$  depend only on the row, respectively only on the column; they contribute diagonal unitary factors on either side and do not affect unitarity. The genuine pairing between row and column is  $m_{12}xv + m_{21}yu$ , of the form (1) with matrix  $\begin{pmatrix} 0 & m_{12} \\ m_{21} & 0 \end{pmatrix}$ , whose determinant is  $-m_{12}m_{21}$ . So  $U^R$  is unitary iff  $m_{12}m_{21}$  is a unit.

*Unitarity of  $U^\Gamma$ .* Reading  $E$  with row index  $(x, v)$  and column index  $(u, y)$ , the terms  $m_{12}xv$  and  $m_{21}yu$  are row-only and column-only respectively, and the pairing matrix is  $\text{diag}(m_{11}, m_{22})$  with determinant  $m_{11}m_{22}$ . So  $U^\Gamma$  is unitary iff  $m_{11}m_{22}$  is a unit.  $\square$

**Theorem 7** (Parity obstruction). *A matrix  $M \in M_2(\mathbb{Z}/d)$  with  $U^{(M)}$  2-unitary exists if and only if  $d$  is odd. For odd  $d$  one may take  $M = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ .*

*Proof.* Suppose  $d$  is even. Every unit of  $\mathbb{Z}/d$  is odd. By Lemma 6 both  $m_{11}m_{22}$  and  $m_{12}m_{21}$  are units, hence odd; therefore  $\det M = m_{11}m_{22} - m_{12}m_{21}$  is a difference of two odd numbers, hence even, hence not a unit of  $\mathbb{Z}/d$ . This contradicts Lemma 6.

If  $d$  is odd, take  $M = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ : then  $m_{11}m_{22} = 2$ ,  $m_{12}m_{21} = 1$  and  $\det M = 1$ , and 2 is a unit modulo any odd  $d$ .  $\square$

We verified Theorem 7 by exhaustive search over all  $d^4$  matrices for  $2 \leq d \leq 12$ ; Table 1 records the census. Note that the obstruction is insensitive to primality: it holds at  $d = 4, 8, 10, 12$  as well as at  $d = 6$ , and *fails* to obstruct at the odd composites  $d = 9, 15$ , where flat bilinear 2-unitaries do exist.

| $d$ | parity | $\#\{M \text{ admissible}\}$ | canonical solution                      |
|-----|--------|------------------------------|-----------------------------------------|
| 2   | even   | 0                            | —                                       |
| 3   | odd    | 8                            | $(1, 1; 1, 2)$ verified 2-unitary, flat |
| 4   | even   | 0                            | —                                       |
| 5   | odd    | 192                          | $(1, 1; 1, 2)$ verified                 |
| 6   | even   | <b>0</b>                     | —                                       |
| 7   | odd    | 1080                         | $(1, 1; 1, 2)$ verified                 |
| 8   | even   | 0                            | —                                       |
| 9   | odd    | 648                          | $(1, 1; 1, 2)$ verified                 |
| 10  | even   | 0                            | —                                       |
| 11  | odd    | 9000                         | $(1, 1; 1, 2)$ verified                 |
| 12  | even   | 0                            | —                                       |

Table 1: Exhaustive census of bilinear phase forms  $M \in M_2(\mathbb{Z}/d)$  yielding a flat 2-unitary of order  $d^2$ . Existence holds exactly for odd  $d$ , in agreement with Theorem 7. Admissibility was tested by the criterion of Lemma 6 and confirmed numerically on representatives.

**Corollary 8.** *Any flat 2-unitary of order 36 — in particular any Butson-type  $\text{AME}(4, 6)$  representative — has a phase function that is not a bilinear form over  $\mathbb{Z}/6$ .*

This gives a precise sense in which the  $d = 6$  solutions must be “non-Gaussian”: the quadratic-Gauss-sum mechanism that manufactures perfect tensors at every odd  $d$  is unavailable, and something genuinely different (in the known constructions, biunimodular vectors [8, 9]) must take its place.

#### 4.1 Twisting by $\mathbb{Z}[\omega]$ does not help

A natural attempt to evade Theorem 7 is to change the coefficient ring: replace  $\mathbb{Z}/d$  by the Eisenstein integers and use phases  $\zeta^{\text{Tr}_{\mathbb{Q}(\omega)/\mathbb{Q}}(\alpha X \bar{U})}$  with  $\alpha \in \mathbb{Z}[\omega]$ , in the hope that the Galois action supplies a sign the parity argument cannot see. It does not, for the following reason.

First, there is no Eisenstein object of size 6 at all: since 2 is inert in  $\mathbb{Z}[\omega]$  its prime ideal has norm 4, so 6 is not the norm of any ideal and no quotient of  $\mathbb{Z}[\omega]$  has six elements. The natural object of size 36 is  $\mathbb{Z}[\omega]/(6)$ , whose additive group is  $\mathbb{Z}/6 \oplus \mathbb{Z}/6$  — which is exactly the index set already in play.

**Proposition 9.** *Write  $X = x + y\omega$  and  $U = u + v\omega$  in  $\mathbb{Z}[\omega]/(6)$ , and  $\alpha = a + b\omega$ . Then*

$$\text{Tr}_{\mathbb{Q}(\omega)/\mathbb{Q}}(\alpha X \bar{U}) = s x u - (s + t) x v + t y u + s y v, \quad s = 2a - b, \quad t = -a - b,$$

*so the trace form is a  $\mathbb{Z}$ -bilinear form with matrix  $M = \begin{pmatrix} s & -s-t \\ t & s \end{pmatrix}$ . Consequently the Eisenstein-twisted family is a subfamily of the bilinear family of Theorem 7: as  $\alpha$  ranges over the 36 classes of  $\mathbb{Z}[\omega]/(6)$  only 12 distinct matrices  $M$  arise, out of the 1296 elements of  $M_2(\mathbb{Z}/6)$ , and none is 2-unitary.*

*Proof.* The displayed identity is a direct expansion using  $\bar{\omega} = -1 - \omega$  and  $\text{Tr}(c + e\omega) = 2c - e$ . For the second claim apply Lemma 6:  $M_{11}M_{22} = s^2$  must be a unit, forcing  $s$  odd;  $M_{12}M_{21} = -(s+t)t$  must be a unit, forcing both  $t$  and  $s+t$  odd. But  $s$  and  $t$  odd makes  $s+t$  even, a contradiction.  $\square$

*Remark 10.* The failure is instructive in two ways. It occurs in the *reshuffling* condition, not the determinant:  $\det M = s^2 + st + t^2$  is the Eisenstein norm form, and for  $s, t$  both odd it is a unit modulo 2, so the determinant condition survives and only  $M_{12}M_{21}$  breaks. Moreover the twist is strictly *worse* than the untwisted construction. Running the criterion over the family for small  $d$  gives passing counts 0, 12, 18, 0, 90 for  $d = 3, 5, 7, 9, 11$ : the Eisenstein trace form fails not only

at even  $d$  but also at  $d = 3$  and  $d = 9$ , because at the ramified prime the conditions “det a unit” and “ $s + t$  a unit” become mutually exclusive. The twisted family therefore requires  $\gcd(d, 6) = 1$ , whereas the plain bilinear family of Theorem 7 works at *every* odd  $d$ .

*Remark 11* (Why no change of ring can work: Hall–Paige). Proposition 9 is a special case of a structural fact. A *complete mapping* of a finite group  $G$  is a bijection  $\theta : G \rightarrow G$  with  $g \mapsto g\theta(g)$  also a bijection; complete mappings are exactly what produce an orthogonal mate for a Cayley table. Hall and Paige proved the elementary direction that a group with nontrivial cyclic Sylow 2-subgroup admits none. There are only two groups of order six,  $\mathbb{Z}/6$  and  $S_3$ , and both have Sylow 2-subgroup  $\mathbb{Z}/2$ : cyclic and nontrivial. We confirm by exhaustive search over all 720 bijections that neither admits a complete mapping.

For cyclic groups the mechanism is literally the parity argument of Theorem 7: a linear complete mapping  $x \mapsto cx$  of  $\mathbb{Z}/n$  requires  $c$  and  $c + 1$  both to be units, which is impossible for even  $n$  since consecutive integers have opposite parity — exactly as  $M_{11}M_{22}$  and  $M_{12}M_{21}$  cannot both be odd while their difference is a unit. Counting linear complete mappings of  $\mathbb{Z}/n$  for  $n = 2, \dots, 12$  gives 0, 1, 0, 3, 0, 5, 0, 3, 0, 9, 0: nonzero precisely for odd  $n$ , matching Table 1 exactly.

The consequence is that the obstruction at six lives in the additive *group* of order six, not in any coefficient ring built on top of it. Replacing  $\mathbb{Z}/6$  by  $\mathbb{Z}[\omega]$ , by  $\mathbb{Z}[i]$ , or by any other ring leaves the group of order six untouched, and both such groups are excluded. This closes off the entire class of “change the ring” constructions at once, and is the conceptual reason the known  $\text{AME}(4, 6)$  representatives are biunimodular rather than character-theoretic.

## 4.2 The obstruction is about the group, not the number

Remark 11 suggests that Theorem 7 is not really a statement about the parity of  $d$  but about the cyclic group  $\mathbb{Z}/d$ . That is correct, and making it precise both generalizes the theorem and shows it is sharp in an unexpected way.

**Definition 12.** A *linear orthomorphism* of a finite abelian group  $A$  is an automorphism  $\sigma \in \text{Aut}(A)$  such that  $\sigma - 1 \in \text{Aut}(A)$ .

Let  $A$  be finite abelian with a perfect pairing  $\langle \cdot, \cdot \rangle : A \times A \rightarrow \mathbb{C}^\times$ , and for  $M \in M_2(\text{End } A)$  set

$$U_{(x,y),(u,v)}^{(M)} = \frac{1}{|A|} \langle x, M_{11}u + M_{12}v \rangle \langle y, M_{21}u + M_{22}v \rangle.$$

**Proposition 13.**  $U^{(M)}$  is 2-unitary if and only if  $M$ ,  $M_{11}M_{22}$  and  $M_{12}M_{21}$  are all automorphisms. Since  $A$  is finite, each  $M_{ij}$  is then itself an automorphism, and after relabelling the four legs by automorphisms — which preserves 2-unitarity — one may take  $M = \begin{pmatrix} 1 & 1 \\ 1 & \sigma \end{pmatrix}$ . Hence a flat bilinear 2-unitary indexed by  $A$  exists if and only if  $A$  admits a linear orthomorphism.

*Proof.* The three conditions are the character-sum computation of Lemma 6 verbatim, with  $\mathbb{Z}/d$  replaced by  $A$  and units by automorphisms; that  $ab \in \text{Aut}(A)$  forces  $a, b \in \text{Aut}(A)$  holds because a surjective endomorphism of a finite group is bijective. For the reduced form, invertibility of  $\begin{pmatrix} 1 & 1 \\ 1 & \sigma \end{pmatrix}$  is equivalent to invertibility of the Schur complement  $\sigma - 1$ .  $\square$

For  $A = \mathbb{Z}/d$  automorphisms are units and  $\sigma - 1$  is  $c - 1$ , so Proposition 13 recovers Theorem 7: consecutive integers have opposite parity, so no linear orthomorphism exists for even  $d$ . The point is that this is a fact about  $\mathbb{Z}/d$ , not about  $d$ .

**Proposition 14.** Take  $A = (\mathbb{Z}/2)^2$  and  $\sigma = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \in GL_2(\mathbb{F}_2)$ . Then  $\sigma$  and  $\sigma - 1 = \sigma + 1 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$  are both invertible, and the resulting  $U^{(M)}$  of order 16 is a flat 2-unitary: all six bipartition purities equal  $1/16$ . Thus there is a bilinear  $\text{AME}(4, 4)$  over the Klein group, although Table 1 records that no bilinear 2-unitary exists over  $\mathbb{Z}/4$ . The same recipe over  $(\mathbb{Z}/2)^3$ , with  $\sigma$  the companion matrix of  $x^3 + x^2 + 1$ , gives a flat bilinear  $\text{AME}(4, 8)$ .

So the even cases split. What fails at  $d = 4$  and  $d = 8$  is the *cyclic* group, not the dimension; replacing it by an elementary abelian 2-group repairs the construction. At  $d = 6$  no such repair exists, because there is no group of order six with non-cyclic Sylow 2-subgroup — indeed no group of order six other than  $\mathbb{Z}/6$  and  $S_3$  at all.

**Corollary 15.** *Some abelian group of order  $d$  admits a linear orthomorphism if and only if  $d \not\equiv 2 \pmod{4}$ .*

*Proof.* If  $d \equiv 2 \pmod{4}$  every group of order  $d$  has Sylow 2-subgroup  $\mathbb{Z}/2$ . Writing  $A = \mathbb{Z}/2 \times B$  with  $B$  of odd order, any  $\sigma$  acts on the  $\mathbb{Z}/2$  factor as the identity, so  $\sigma - 1$  kills it and is not injective. Conversely for odd  $d$  take  $\sigma = 2$ , and for  $4 \mid d$  write the Sylow 2-part with a  $(\mathbb{Z}/2)^2$  or larger elementary factor and use Proposition 14 componentwise.  $\square$

*Remark 16* (Linear is strictly stronger than Hall–Paige). Linear orthomorphisms are rarer than complete mappings. The smallest witness is  $A = \mathbb{Z}/4 \times \mathbb{Z}/2$ : its Sylow 2-subgroup is non-cyclic, so by Hall–Paige it *does* admit a complete mapping, yet an exhaustive search over  $\text{End}(A)$  finds no linear orthomorphism. The reason is clean: reduction modulo  $2A$  gives  $A/2A \cong \mathbb{F}_2^2$ , and a linear orthomorphism would induce  $\bar{\sigma}$  with  $\bar{\sigma}, \bar{\sigma} + 1 \in GL_2(\mathbb{F}_2)$ , forcing  $\bar{\sigma}$  to have order 3; but  $|\text{Aut}(\mathbb{Z}/4 \times \mathbb{Z}/2)| = 8$  has no element of order 3. So “character-theoretic” is a strictly narrower notion than “group-based”.

*Remark 17* (Four tiers, with six alone at the bottom). Collecting what is known, 2-unitaries of order  $d^2$  fall into four regimes:

| case                             | OLS( $d$ ) | group-based | mechanism                                        |
|----------------------------------|------------|-------------|--------------------------------------------------|
| $d$ odd                          | yes        | yes         | finite field / bilinear                          |
| $d \equiv 0 \pmod{4}$            | yes        | yes         | elementary abelian 2-group; $\mathbb{Z}/d$ fails |
| $d \equiv 2 \pmod{4}, d \geq 10$ | yes        | <b>no</b>   | Bose–Shrikhande–Parker; no admissible group      |
| $d = 6$                          | <b>no</b>  | <b>no</b>   | entangled solutions only                         |
| $d = 2$                          | <b>no</b>  | <b>no</b>   | no AME(4, 2) exists [11]                         |

The third row is the interesting one: for every  $d \equiv 2 \pmod{4}$  each group of order  $d$  has cyclic Sylow 2-subgroup, so Hall–Paige forbids a complete mapping and no Cayley-table construction can succeed — yet from  $d = 10$  onwards orthogonal Latin squares exist anyway, by [2]. Euler’s conjecture that OLS( $d$ ) fails for all  $d \equiv 2 \pmod{4}$  is thus *true* for group-based squares and false in general; the Bose–Shrikhande–Parker counterexamples are necessarily non-group. Dimension six is the unique place where the combinatorial object fails as well, leaving only the entangled solutions of [4] and [7].

## 5 Anatomy of two AME(4, 6) representatives

### 5.1 The golden representative

We parsed the analytic form of the golden AME(4, 6) matrix from the authors’ own source code and re-verified it independently. It has 112 nonzero entries with three amplitude values

$$a = \frac{1}{\sqrt{5+\sqrt{5}}}, \quad b = \frac{\sqrt{5+\sqrt{5}}}{\sqrt{20}}, \quad c = \frac{1}{\sqrt{2}},$$

and phases in  $\mu_{20}$ ,  $w = e^{i\pi/10}$ . We confirm  $b/a = \varphi$  exactly and  $a^2 + b^2 = c^2 = \frac{1}{2}$ ; that  $U$ ,  $U^R$ ,  $U^\Gamma$  are unitary; and that all six bipartition purities equal  $1/36$  to machine precision. The amplitude census is 40 entries of modulus  $a$ , 40 of modulus  $b$ , and 32 of modulus  $c$ .

Motivated by the exceptional 2-transitive action of  $\text{PGL}_2(\mathbb{F}_5) \supset \text{PSL}_2(\mathbb{F}_5) \cong A_5$  on the six points of  $\mathbb{P}^1(\mathbb{F}_5)$  — the source of the outer automorphism of  $S_6$ , and of golden character values — it is natural to ask whether the golden state is covariant for a copy of  $A_5$  acting on the six levels. We tested this exhaustively.

**Proposition 18.** *Let  $\mathcal{E} \subset (\mathbb{Z}/6)^4$  be the support of the golden representative, colored by amplitude class. The group of quadruples  $(\pi_1, \pi_2, \pi_3, \pi_4) \in S_6^4$  acting independently on the four legs and preserving  $\mathcal{E}$  together with its coloring has order 2. Its nontrivial element is*

$$\pi_1 = (0\ 1)(2\ 3)(4\ 5), \quad \pi_2 = (0\ 5)(1\ 4)(2\ 3), \quad \pi_3 = \pi_4 = \text{id}.$$

*Moreover this element does not lift to a monomial symmetry: there are no diagonal unitaries  $D_L, D_R$  with  $D_L(P_{\pi_1} \otimes P_{\pi_2}) U D_R = U$ .*

*Proof (computational).* Leg-wise color profiles reduce the search to 48, 48, 4, 8 candidates on the four legs; pair-profile filtering and constraint propagation on the fibers leave 2 quadruples, both of which are then verified to preserve all 112 colored edges. For the second claim, write  $V = (P_{\pi_1} \otimes P_{\pi_2})U$ ; then  $|V| = |U|$ , and a monomial lift exists iff the phase ratio  $\arg(V_{jk}/U_{jk})$  is of the separable form  $r_j + c_k$  on the support. The support bipartite graph has 9 connected components, each of size 8; performing the propagation *within each component* (the assignment is free between components) yields residuals up to 2.51 radians, so no separable solution exists.  $\square$

**Corollary 19.** *The golden representative is not covariant for any nontrivial group of level permutations acting monomially; in particular it carries no  $A_5$  or  $\text{PGL}_2(\mathbb{F}_5)$  symmetry.*

*Remark 20 (Scope).* This is a statement about *this representative*. AME states are defined up to local unitaries, and a symmetry invisible in one frame may exist in another; the negative result rules out permutation-monomial symmetry in the analytic frame of [4], not in the whole LU class. We flag the possibility that the golden solution’s arithmetic  $(\mu_{20}, \varphi)$  is a feature of the seed — the algorithm of [4] is initialized near the best classical permutation [14] — rather than of the state.

The next proposition explains where  $\varphi$  does come from, and is consistent with Proposition 18: the golden ratio is a *local* datum of the  $2 \times 2$  blocks, not a shadow of a global icosahedral symmetry.

**Proposition 21.** *The three amplitudes of the golden representative satisfy the closed forms*

$$a^2 = \frac{1}{2\varphi\sqrt{5}}, \quad b^2 = \frac{\varphi}{2\sqrt{5}}, \quad c^2 = \frac{1}{2},$$

*so that  $a^2 + b^2 = c^2$  and  $b/a = \varphi$ . Equivalently, the within-block probability split is*

$$\left(\frac{1}{\varphi\sqrt{5}}, \frac{\varphi}{\sqrt{5}}\right) = (0.2763932\dots, 0.7236067\dots),$$

*which sums to 1 by the identity  $\varphi^2 + 1 = \varphi\sqrt{5}$ , and the normalized block vector is*

$$\frac{(a, b)}{c} = \frac{(1, \varphi)}{\sqrt{2} + \varphi} = (0.5257311\dots, 0.8506508\dots),$$

*the pair of nonzero coordinates of a vertex of the regular icosahedron. Moreover this differs from the normalized column  $(1/\varphi, 1/\sqrt{\varphi}) = (0.6180339\dots, 0.7861513\dots)$  of the Fibonacci  $F$ -matrix, which is normalized by the distinct golden identity  $\varphi^{-1} + \varphi^{-2} = 1$ .*

*Remark 22.* Thus the golden amplitudes are *not* Fibonacci fusion data, and the two natural “golden splits”  $(\varphi/\sqrt{5}, 1/(\varphi\sqrt{5}))$  and  $(\varphi^{-1}, \varphi^{-2})$  must not be conflated. Combined with Proposition 18, the picture is:  $\varphi$  appears as the icosahedral vertex ratio inside each block, and does not propagate to a symmetry of the 36-officer array.



## 5.2 Block structure, and what the local datum actually is

The support of the golden representative is highly disconnected, and the components are the natural carriers of its arithmetic.

**Proposition 23.** *After row and column permutation the golden representative is block diagonal with exactly 9 blocks, each a unitary  $4 \times 4$  matrix. Eight of them are mutually equivalent, with 12 nonzero entries and every row of squared moduli a permutation of*

$$(0, a^2, b^2, c^2) = (0, 0.1381966\dots, 0.3618033\dots, 0.5).$$

*The ninth is exceptional: it has full support, 16 nonzero entries, and every row of squared moduli a permutation of  $(a^2, a^2, b^2, b^2)$ . Total support  $8 \cdot 12 + 16 = 112$ .*

**Proposition 24** (Support asymmetry). *The support of the golden representative is not symmetric under transposition. Its row and column support profiles are*

$$\text{rows: } 32 \times 3 + 4 \times 4, \quad \text{columns: } 16 \times 2 + 20 \times 4,$$

*both summing to 112. The asymmetry is entirely accounted for by Proposition 23: each generic block has row supports  $(3, 3, 3, 3)$  and column supports  $(2, 2, 4, 4)$ , contributing 16 columns of support 2; the exceptional block contributes four more columns of support 4. Moreover, reshaping each line as a  $6 \times 6$  matrix, every one of the 36 columns has Schmidt rank exactly 2, while the rows have Schmidt rank 3 (32 of them) or 4 (4 of them).*

*Remark 25* (Which quantities are invariants). Several natural-looking quantities attached to a block are not well defined. The blocks of Proposition 23 are extracted by independent row and column index sets, so there is no canonical pairing of rows with columns; consequently the *eigenvalues* of a block are not invariants. Concretely, transposing two rows of the exceptional block changes its spectrum from  $\{-0.979 - 0.206i, 0.486 - 0.874i, 0.604 + 0.797i, 0.929 - 0.371i\}$  to  $\{-0.246 - 0.969i, -0.178 + 0.984i, 0.660 - 0.751i, 0.822 + 0.569i\}$ . The quantities that are invariant, and which we use, are the singular values (all equal to 1, since each block is unitary) and the multiset of entry moduli, which is what Corollary 27 rests on.

**Proposition 26.** *The block amplitudes admit the parametrization*

$$a^2 = \frac{1}{4} \left( 1 - \frac{1}{\sqrt{5}} \right), \quad b^2 = \frac{1}{4} \left( 1 + \frac{1}{\sqrt{5}} \right),$$

*so that, with  $\theta = \arctan \varphi$  the block angle,  $\cos 2\theta = -1/\sqrt{5}$  and  $\sin 2\theta = 2/\sqrt{5}$ . Hence the primitive datum of the block is  $\sqrt{5}$ , the quadratic Gauss sum of conductor 5, and  $\varphi = b/a$  is derived from it by the half-angle identity  $b/a = \sqrt{(\sqrt{5} + 1)/(\sqrt{5} - 1)} = \varphi$ .*

This settles, at the level of the  $2 \times 2$  datum, the question of whether the golden block is Fibonacci fusion data.

**Corollary 27.** *No block of the golden representative is gauge-equivalent to the Fibonacci  $F$ -matrix.*

*Proof.* Gauge transformations act by  $F \mapsto D_1 F D_2$  with  $D_i$  diagonal unitary, together with relabelling; the multiset of entry moduli is therefore a gauge invariant. The normalized golden datum has moduli  $\{1/\sqrt{2+\varphi}, \varphi/\sqrt{2+\varphi}\} = \{0.5257311\dots, 0.8506508\dots\}$ , while the Fibonacci  $F$ -matrix has moduli  $\{1/\varphi, 1/\sqrt{\varphi}\} = \{0.6180339\dots, 0.7861513\dots\}$ . The two multisets differ, and the underlying normalizations are the distinct golden identities  $1 + \varphi^2 = \varphi\sqrt{5}$  and  $\varphi^{-1} + \varphi^{-2} = 1$ .  $\square$

*Remark 28.* The broader question — whether some block is an  $F$ -matrix of *some* fusion category — is not settled by Corollary 27, but is now sharply constrained: any such category must reproduce the row profile  $(0, a^2, b^2, c^2)$  or  $(a^2, a^2, b^2, b^2)$ . We also record a negative result guarding against an appealing guess: although the exceptional block’s modulus pattern coincides with that of a Kronecker product  $R \otimes H$  ( $R$  the  $2 \times 2$  icosahedral rotation,  $H$  a flat  $2 \times 2$  unitary), an exhaustive search over all  $4! \times 4!$  row and column permutations shows that it does *not* factor as such a product once phases are taken into account. Matching modulus patterns do not imply tensor structure.

*Remark 29* (Both families are quadratic Gauss sums at odd conductor). Proposition 26 places the two known families of  $\text{AME}(4, 6)$  representatives in a common frame. The golden family has coefficient field  $\mathbb{Q}(\zeta_{20})$ , which contains the conductor-5 quadratic Gauss sum  $\sum_a (\frac{a}{5}) \zeta_5^a = \sqrt{5}$ ; the Butson family of Proposition 31 has coefficient field  $\mathbb{Q}(\zeta_6)$ , which contains the conductor-3 Gauss sum  $\sqrt{-3}$ . Both are thus built over the quadratic Gauss sum of an odd prime conductor — the two smallest — while Theorem 7 says the even part of the modulus can supply no such construction at all. We state this as an observation about the two known solutions, not as a classification: by [6] the moduli space is infinite, and we have examined two points of it.

It is tempting to separate the golden and Butson representatives by their support density  $F_1 := \#\{\text{nonzero entries}\}/1296$ : one computes

$$F_1(\text{golden}) = \frac{112}{1296} = \frac{7}{81} = 0.0864\dots, \quad F_1(\text{Butson}) = 1.$$

This separation is real but must not be read as a statement about the  $\text{AME}(4, 6)$  moduli space, because  $F_1$  is not invariant under local unitaries: applying a generic  $V_1 \otimes V_2 \otimes V_3 \otimes V_4$  to the golden state leaves the state’s LU class unchanged while sending  $F_1$  to 1. The corrected functional is the orbit minimum  $F_1^*(\psi) := \min\{F_1(U) : U \in \text{LU-orbit of } \psi\}$ . The following pins down its range.

**Proposition 30.** *Every 2-unitary of order 36 has at least 37 nonzero entries. Consequently*

$$\frac{1}{36} < F_1^* \leq \frac{7}{81}$$

*for the golden state.*

*Proof.* Suppose  $U$  is 2-unitary with exactly 36 nonzero entries. Since  $U$  is unitary every row and every column has norm 1, hence at least one nonzero entry; with only 36 entries available among 36 rows and 36 columns, each row and each column carries exactly one. So  $U$  is monomial,  $U = DP$  with  $D$  diagonal unitary and  $P$  a permutation matrix; write  $T_{jklm} \in \{0, 1\}$  for the 0/1 tensor of  $P$ , so that  $|U_{jk;lm}| = T_{jklm}$ .

Unitarity of  $U^R$  forces each of its rows to have norm 1: the row indexed by  $(j, l)$  has entries  $U_{jk;lm}$  as  $(k, m)$  varies, so  $\sum_{k,m} T_{jklm} = 1$  for every  $(j, l)$ . Likewise unitarity of  $U^T$  gives  $\sum_{k,l} T_{jklm} = 1$  for every  $(j, m)$ , and unitarity of  $U$  gives  $\sum_{l,m} T_{jklm} = 1$  for every  $(j, k)$ . These are exactly the conditions under which  $T$  encodes a pair of orthogonal Latin squares of order 6 [10]; the diagonal phases  $D$  play no role in the counting. No such pair exists [1, 3], a contradiction.  $\square$

Whether the golden value  $7/81$  is itself optimal — equivalently, whether 112 is the minimum support of any  $\text{AME}(4, 6)$  representative — we leave open.

### 5.3 The Butson (Eisenstein) representatives

We rebuilt the biunimodular construction of [7, 8] from scratch: with  $F_6$  the Fourier matrix,  $P$  the mod-6 generalized CNOT, and  $K = (F_6 \otimes \mathbb{I})P(F_6 \otimes \mathbb{I})$ ,  $L = (F_6 \otimes \mathbb{I})P(F_6^\dagger \otimes \mathbb{I})$ , we form  $\mathcal{U}_j = K \text{diag}(e^{2\pi i \Lambda_j}) L$ . We verify  $K = K^\dagger$ ,  $L = L^T$ , and that both are flat.

**Proposition 31.**  *$\mathcal{U}_1$  and  $\mathcal{U}_2$  are flat 2-unitary matrices of order 36: all entries have modulus  $1/6$ , all of  $\mathcal{U}$ ,  $\mathcal{U}^R$ ,  $\mathcal{U}^T$  are unitary, and all six bipartition purities equal  $1/36$ . Every entry is exactly a sixth root of unity: writing  $\mathcal{U}_{jk} = \frac{1}{6}\zeta_6^{m_{jk}}$  with  $m_{jk} \in \mathbb{Z}/6$ , the reconstruction error is  $8.3 \times 10^{-15}$ . Hence  $6\mathcal{U}_j \in \text{BH}(36, 6)$  and all phases lie in  $\mathbb{Z}[\omega]^\times$ .*

For  $\mathcal{U}_1$  the exponent census over  $\mathbb{Z}/6$  is  $(204, 180, 294, 180, 204, 234)$  for  $m = 0, \dots, 5$ . We then probed the structure of the phase function  $m(x, y, u, v)$  on  $(\mathbb{Z}/6)^4$ .

*Observation 32.* For  $\mathcal{U}_1$ :

- (a) *Non-polynomiality.*  $m$  is not a polynomial of total degree  $\leq 2$  over  $\mathbb{Z}/6$ : fitting the 15 monomials of degree  $\leq 2$  in four variables against all 1296 points is inconsistent both modulo 2 (rank 11) and modulo 3 (rank 15).
- (b) *No translation covariance.* None of the shifts  $(x, u) \mapsto (x + t, u + t)$ ,  $(y, v) \mapsto (y + t, v + t)$ , or a simultaneous shift of all four legs changes  $m$  by a constant, for any  $t = 1, \dots, 5$ .
- (c) *Distance from bilinearity.* The best pure bilinear form agrees with  $m$  on 372 of 1296 points (28.7%, against a 16.7% random baseline). The maximum is attained at more than one  $M$  (e.g.  $M = (2, 3; 1, 2)$ ); only the value 372 is canonical.
- (d) *CRT non-factorization.* Write  $\zeta_6^m = (-1)^m \bar{\omega}^m$ , splitting the phase matrix into a  $\chi_2$ -layer  $S \in \{\pm 1\}^{36 \times 36}$  and a  $\chi_3$ -layer  $W \in \mu_3^{36 \times 36}$ . Neither layer is unitary after normalization: the Gram spectrum of  $S/6$  runs from 0 (so  $S$  is singular) to 9, and that of  $W/6$  from  $1/3$  to 7.

Part (d) is the structurally sharpest: the solution cannot be assembled from a real Hadamard factor and an Eisenstein factor separately. Together with Theorem 7 it says that the 2-adic and 3-adic layers of  $\mathbb{Z}/6$  are irreducibly entangled in any flat order-36 solution — the same factorization  $6 = 2 \cdot 3$  that destroys the field  $\mathbb{F}_6$  and hence the bilinear route also forbids the solution from splitting along that factorization. Parts (a)–(c) confirm that the phase function is genuinely of “biunimodular” rather than Gauss-sum type.

*Remark 33.* Our transcription of the third vector  $\Lambda_3$  from the published PDF failed its own biunimodularity test; we attribute this to text-extraction corruption on our side rather than to an error in [7], and have therefore excluded  $\mathcal{U}_3$  from Proposition 31. The authors’ machine-readable data is available at their repository.

## 6 A companion arithmetic thread: weight 36 and $\chi_{-3}$

The objects of Sections 3–5 are governed by the modulus 6 and the field  $\mathbb{Q}(\sqrt{-3})$ . The same two data organize a classical arithmetic thread in which the number 36 reappears as a *weight* rather than a dimension. We record here what we could verify, and state plainly that we have no proven bridge between the two threads: the shared modulus is a thematic link, not a theorem.

Throughout,  $\chi_{-3}$  denotes the odd quadratic character modulo 3, i.e. the Kronecker symbol  $\left(\frac{-3}{\cdot}\right)$ . This is the character in the Eisenstein master identity  $r(n)/6 = \sum_{d|n} \chi_{-3}(d)$  for representations by  $x^2 + xy + y^2$ , and the non-principal character modulo 6 is induced by it; the two  $L$ -functions differ only by the Euler factor at 2,  $L(s, \chi_6) = L(s, \chi_{-3})(1 + 2^{-s})$ .

### 6.1 Degree of the modular-zero polynomial

Let  $k = 12m$ , so  $\dim S_k = m$ , and let  $f_1, \dots, f_m$  be the normalized Hecke eigenforms. Define the *modular-zero polynomial*  $P_k(X) = \prod_i \prod_\tau (X - j(\tau))$ , the product over eigenforms and over their zeros  $\tau$  in the open upper half-plane modulo  $\mathrm{SL}_2(\mathbb{Z})$ .

**Proposition 34.** *Generically*  $\deg P_{12m} = m(m - 1)$ .

*Proof.* A normalized eigenform has  $a_1 = 1$ , hence  $\mathrm{ord}_\infty f_i = 1$  exactly. The valence formula

$$\mathrm{ord}_\infty f + \frac{1}{2}\mathrm{ord}_i f + \frac{1}{3}\mathrm{ord}_\rho f + \sum_{P \neq i, \rho} \mathrm{ord}_P f = \frac{k}{12} = m$$

therefore leaves total order  $m - 1$  among the points of the upper half-plane. Away from the elliptic points these are  $m - 1$  zeros contributing  $m - 1$  distinct values of  $j$ ; summing over the  $m$  eigenforms gives  $m(m - 1)$ . (Zeros at  $i$  or  $\rho$ , which carry weights  $\frac{1}{2}$  and  $\frac{1}{3}$ , would alter the count; the statement is generic in that sense.)  $\square$

The formula gives 0, 2, 6, 12, 20 for  $m = 1, \dots, 5$ . The case  $m = 1$  is a consistency check:  $\Delta$  has its only zero at the cusp, so  $P_{12} = 1$ . The value 6 at  $m = 3$  matches the degree of the weight-36 polynomial obtained previously as a symbolic resultant. An independent computation reports irreducibility for  $m = 2, 3, 4, 5$  and  $\text{sign}(\text{disc}) = +, -, -, +$  respectively.

## 6.2 The weight-36 Hecke cubic

We computed  $S_{36}$  from scratch in the echelon basis  $\Delta E_4^6, \Delta^2 E_4^3, \Delta^3$  (leading terms  $q, q^2, q^3$ ) and applied  $T_2 f = \sum_n (a(2n) + 2^{35} a(n/2)) q^n$ .

**Proposition 35.** *In that basis*

$$T_2 = \begin{pmatrix} 1416 & 1 & 0 \\ 34629120000 & 144384 & -72 \\ -1791590400000 & -331776000 & -6144 \end{pmatrix},$$

with characteristic polynomial

$$\chi(x) = x^3 - 139656x^2 - 59208339456x - 1467625047588864.$$

This cubic is irreducible over  $\mathbb{Q}$  and totally real, with roots  $-165109.1678\dots, -26808.0076\dots, 331573.1754\dots$ , and

$$\text{disc } \chi = 2^{30} \cdot 3^{10} \cdot 5^2 \cdot 7^2 \cdot 23 \cdot 1259 \cdot \mathbf{269461929553},$$

which is not a square; hence  $\text{Gal}(\chi/\mathbb{Q}) \cong S_3$  and the Hecke field is a totally real non-Galois cubic.

*Remark 36* (Corroboration across independent computations). An independent computation communicated to us reports, for the degree-6 field  $\mathbb{Q}(j_0)$  generated by a modular zero in weight 36: Galois group  $C_2^3 \rtimes S_3 = 2 \wr S_3$  of order 48 (imprimitive, permuting three blocks of two — one conjugate pair per eigenform), signature  $(r_1, r_2) = (4, 1)$ , class number  $h_K = 16$  with  $\text{Cl}_K \cong C_2^4$ , and field discriminant

$$-3 \cdot 5^2 \cdot 7^2 \cdot 23^2 \cdot 233 \cdot 269 \cdot 1259^2 \cdot \mathbf{269461929553}^2.$$

We did not reproduce the class-group data, having no class-group software available. We do note that our independently computed  $\text{disc } \chi$  shares with it the primes 3, 5, 7, 23, 1259 and the 12-digit prime 269461929553, with the odd-exponent primes appearing squared upstairs — precisely the behaviour expected of a quadratic-type extension of the cubic subfield. Two computations performed by different routes agreeing on a 12-digit discriminant prime is strong mutual corroboration. Consistently, the negative discriminant matches signature  $(4, 1)$  via  $\text{sign} = (-1)^{r_2}$ .

*Remark 37* (The sextic symbol forces the modulus 6). One further appearance of 6 in the Eisenstein field deserves recording, since it is forced rather than chosen. A degree- $m$  residue symbol  $\left(\frac{a}{\varpi}\right)_m \equiv a^{(N(\varpi)-1)/m}$  is defined only at prime ideals with  $N(\varpi) \equiv 1 \pmod{m}$ . Over  $\mathbb{Z}[i]$  one has  $\mu_4 \subset \mathbb{Z}[i]$ , and both the quartic and quadratic symbols require removing only the ramified prime  $(1 + i)$ , of norm 2. Over  $\mathbb{Z}[\omega]$  one has  $\mu_6 = \mathbb{Z}[\omega]^\times$ , so a sextic symbol exists; but  $N(\varpi) \equiv 1 \pmod{6}$  fails exactly at  $(1 - \omega)$ , of norm 3, and at the prime above 2, of norm 4. Hence the sextic symbol — which exists precisely because the unit group has order six — is defined exactly on the prime ideals coprime to 6. The unit group of order 6 and the modulus 6 are the same 6 here, and neither is imposed: the size of  $\mathbb{Z}[\omega]^\times$  determines the coprimality condition. The same  $\mu_6$  carries the phases of the Butson representatives of Proposition 31.

Writing  $L(s, \chi_{-3}) = 3^{-s}(\zeta(s, \frac{1}{3}) - \zeta(s, \frac{2}{3}))$  and forming the Hardy function  $Z(t) = e^{i\theta(t)}L(\frac{1}{2} + it, \chi_{-3})$  with  $\theta(t) = \frac{t}{2} \log \frac{3}{\pi} + \Im \log \Gamma(\frac{3}{4} + \frac{it}{2})$ , we find  $Z$  real to  $10^{-21}$ , confirming root number  $+1$ .

*Observation 38.*  $L(s, \chi_{-3})$  has exactly 34 zeros with  $5 \leq t \leq 80$ , the first eight at

$$8.0397, 11.2492, 15.7046, 18.2620, 20.4558, 24.0594, 26.5779, 28.2182.$$

At the first zero,  $|L(\sigma + it_1)|$  is minimized at  $\sigma = \frac{1}{2}$ , where it equals  $1.9 \times 10^{-16}$ , against  $2.97 \times 10^{-2}$  already at  $\sigma = 0.475$ . The normalized Chebyshev bias satisfies

$$\max_{x \leq 50000} \frac{|\psi(x, \chi_{-3})|}{\sqrt{x}} = 1.0428 \quad \text{at } x = 269.$$

**Lemma 39.** *The normalized bias maximum is the same whether computed with the character modulo 3 or the character modulo 6.*

*Proof.* The two summatory functions differ only on prime powers  $2^k$ . Up to  $x = 269$  there are exactly eight such,  $2, 4, \dots, 256$ , and  $\chi_{-3}(2^k) = (-1)^k$ , so their contributions cancel in pairs and  $\psi(269, \chi_{-3}) = \psi(269, \chi_6)$ .  $\square$

*Remark 40* (What this does and does not establish). Observation 38 is *not* new evidence for GRH at conductor 3, and we correct an earlier framing of our own. Platt [17] verified GRH rigorously, in interval arithmetic, for all primitive Dirichlet  $L$ -functions of modulus  $q \leq 400\,000$  up to height  $\max(10^8/q, 3.75 \cdot 10^7/q + 200)$  for odd  $q$ ; at  $q = 3$  that is height  $\approx 3.3 \times 10^7$ . Our computation re-verifies a range settled long ago. More generally, a finite bound on  $|\psi(x, \chi)|/\sqrt{x}$  cannot exclude zeros off the critical line at all: it yields only a quantitative region  $\beta \leq \frac{1}{2} + c(\gamma)/\log X$ , which approaches but never reaches  $\frac{1}{2}$ . The implication “bounded normalized bias  $\Rightarrow$  GRH” needs the bound for all  $x$ .

What the data can do is test the *distributional* predictions that GRH assumes rather than the hypothesis itself, which is the content of Observation 41.

*Observation 41* (Second moment: a test of GRH and zero simplicity). Under GRH the explicit formula gives  $\psi(x, \chi)/\sqrt{x} = -\sum_{\gamma > 0} 2(\frac{1}{4} + \gamma^2)^{-1/2} \cos(\gamma \log x - \theta_\gamma)$ . For  $\gamma \neq \gamma'$  the logarithmic mean of  $\cos(\gamma y - \theta) \cos(\gamma' y - \theta')$  vanishes by orthogonality alone, so with no independence assumption the logarithmic distribution of  $\psi(x, \chi)/\sqrt{x}$  has mean 0 and variance

$$\sigma^2 = \sum_{\gamma > 0} \frac{2}{\frac{1}{4} + \gamma^2},$$

provided only that the ordinates are distinct, i.e. that the zeros are simple. We locate all 78 zeros with  $0 < \gamma < 150$ ; the count agrees with  $N(T) = \theta(T)/\pi + 1 + S(T)$  with  $|S(T)| < 1.7$  and oscillating in sign, so the list is complete and every zero to that height is on the line and simple. Summing over them gives 0.10202; the tail  $\int_{150}^{\infty} 2t^{-2} dN(t)$  with  $dN(t) = (2\pi)^{-1} \log(qt/2\pi) dt$  contributes 0.01119, so  $\sigma^2 \approx 0.11321$ . The density used for the tail is confirmed sharply by the count itself:  $(T/2\pi) \log(qT/2\pi e) = 78.098$  at  $T = 150$ , against 78 observed. Against this, the empirical logarithmic second moment  $(\int_{X_0}^X (\psi/\sqrt{x})^2 dx/x) / \log(X/X_0)$  with  $X = 10^7$  takes the values 0.1143, 0.1153, 0.1155 for  $X_0 = 10^2, 10^3, 10^4$ , and over dyadic windows  $[2^{k-1}, 2^k)$  it oscillates between 0.057 and 0.158 with no drift.

*Remark 42* (The second moment does *not* test linear independence). It is tempting to describe Observation 41 as a test of GRH together with linear independence of the ordinates over  $\mathbb{Q}$ , and we corrected an earlier draft of our own on this point. Linear independence plays no role: the cross terms die by orthogonality whenever  $\gamma \neq \gamma'$ , and simplicity of the zeros is all that is used. LI first becomes visible in the fourth moment, where nontrivial resonances  $\gamma_1 + \gamma_2 = \gamma_3 + \gamma_4$  would contribute off-diagonal terms that LI forbids.

*Observation 43* (Fourth moment: a test of LI). If the phases do equidistribute independently, then writing  $a_\gamma = 2(\frac{1}{4} + \gamma^2)^{-1/2}$  one has  $\mathbb{E}[S^4] = 3\sigma^4 - \frac{3}{8} \sum_\gamma a_\gamma^4$ , so the logarithmic kurtosis is  $3 - \frac{3}{8}(\sum_\gamma a_\gamma^4)/\sigma^4$ , strictly below the Gaussian value 3. Here  $\sum_\gamma a_\gamma^4 = 16 \sum_\gamma (\frac{1}{4} + \gamma^2)^{-2} = 0.005493$  (the tail beyond 150 is  $1.2 \times 10^{-6}$ ), giving predicted fourth moment 0.03639 and kurtosis 2.8393. Empirically, at  $X = 10^7$  the fourth moment is 0.0351, 0.0361, 0.0359 and the kurtosis 2.686, 2.716, 2.690 for  $X_0 = 10^2, 10^3, 10^4$ . The fourth moment agrees to within 1.5%; the kurtosis gap of about 5% is the same discrepancy amplified, since the empirical  $\sigma^2$  runs 2% high and enters squared. No gross resonance is visible, which is mild positive evidence for LI at this conductor; convergence of fourth moments is slow and we claim nothing sharper.

*Observation 44* (The full limiting distribution). Both moment tests are subsumed by a single sharper one. Since  $\mathbb{E}[e^{ita \cos U}] = J_0(at)$  for  $U$  uniform, GRH + LI predict that the logarithmic distribution of  $\psi(x, \chi)/\sqrt{x}$  has characteristic function

$$\Phi(t) = \prod_{\gamma > 0} J_0(a_\gamma t), \quad a_\gamma = \frac{2}{\sqrt{\frac{1}{4} + \gamma^2}},$$

computable from the zeros with no free parameters; we take the product over the 78 computed zeros and model  $\gamma > 150$  by the Gaussian factor  $e^{-\sigma_{\text{tail}}^2 t^2/2}$  with  $\sigma_{\text{tail}}^2 = 0.01119$ . Numerical inversion gives a density whose moments reproduce the predictions of Observations 41 and 43 to  $10^{-13}$ , confirming the quadrature. Comparing with the empirical distribution on  $[10^3, 10^7]$ :

| $c$                             | 0.25   | 0.50   | 0.75    | 1.00    |
|---------------------------------|--------|--------|---------|---------|
| $\mathbb{P}( S  > c)$ predicted | 0.4661 | 0.1392 | 0.02333 | 0.00187 |
| empirical                       | 0.4815 | 0.1371 | 0.02249 | 0.00099 |

and  $\sup_x |F_{\text{pred}}(x) - F_{\text{emp}}(x)| = 0.034$  over  $[-1.5, 1.5]$ . The agreement holds across two orders of magnitude in probability. This tests all moments simultaneously, and the tails — where LI is most consequential — show no anomaly.

*Observation 45* (Zero repulsion, and the simplicity assumption). Observation 41 presumes the ordinates are distinct. This can be tested directly. Extending to  $T = 300$  gives 189 zeros, and normalising gaps by the local density  $(2\pi)^{-1} \log(q\gamma/2\pi)$  yields mean normalised gap 0.99988 — an independent confirmation that no zero has been missed. The empirical gap distribution follows the GUE surmise rather than the Poisson law:

| $s$           | 0.25  | 0.50  | 0.75  | 1.00  |
|---------------|-------|-------|-------|-------|
| empirical CDF | 0.005 | 0.032 | 0.234 | 0.532 |
| GUE           | 0.016 | 0.112 | 0.302 | 0.533 |
| Poisson       | 0.221 | 0.394 | 0.528 | 0.632 |

The smallest normalised gap observed is 0.085, near  $\gamma \approx 246.3$ ; a double zero would require 0. At small  $s$  the data show slightly more repulsion than GUE (one gap below 0.25 against an expected three), which we attribute to the small sample. Poisson statistics, which permit near-coincident zeros freely, are excluded.

## 7 Discussion and open problems

**A hexit holographic brick.** A verified four-leg perfect tensor of local dimension six is exactly the ingredient required to build a HaPPY-type holographic code [13] on six-dimensional legs. Unlike the qubit pentagon code, such a brick cannot arise from a stabilizer/OLS construction, by Theorem 7 together with the nonexistence of OLS(6). Constructing the associated  $((3, 6, 2))_6$  code explicitly and studying entanglement wedges on hexit legs is the natural next computation.

**The MUB(6) wall.** Theorem 3 exhibits, in dimension 3, an exact coincidence between an Eisenstein lattice structure and a complete MUB system. It is natural to ask whether the Butson representatives of Proposition 31 contain unbiasedness data in dimension six. We have now carried out that test, and the answer is negative.

**Proposition 46.** *Decompose  $\mathcal{U}_1$  and  $\mathcal{U}_2$  into 36 blocks of size  $6 \times 6$ , in either of the two natural ways — contiguous blocks  $U[6j:6j+6, 6l:6l+6]$ , or the reshuffling blocks  $B_{km}^{(j,l)} = U_{jk;lm}$ . Every block has Hilbert–Schmidt norm 1, but no block of either matrix, in either decomposition, is a scaled unitary. Consequently no block is an orthonormal basis of  $\mathbb{C}^6$  and no pair of blocks can be mutually unbiased.*

*Proof (computational).* Since  $\mathcal{U}$  is flat,  $\sqrt{6}B$  has all entries of modulus  $1/\sqrt{6}$ , so  $\sqrt{6}B$  is a scaled unitary precisely when  $6B$  is a complex Hadamard matrix of order 6, equivalently when all six singular values of  $\sqrt{6}B$  equal 1. Computing all 36 spectra in each case, one finds exactly four distinct spectra with multiplicities  $(12, 12, 6, 6)$ , and none is  $(1, 1, 1, 1, 1, 1)$ . In units of  $\sqrt{2/3}$  the spectra are, for  $\mathcal{U}_1$ ,

$$\begin{aligned} (1.803, 1.5, 1.323, .866, .866, .5)^{\times 12}, & \quad (1.803, 1.323, 1.323, 1.323, .5, .5)^{\times 12}, \\ (2, 1, 1, 1, 1, 1)^{\times 6}, & \quad (\sqrt{3}, \sqrt{3}, 1, 1, 1, 0)^{\times 6}, \end{aligned}$$

and correspondingly for  $\mathcal{U}_2$ . □

*Remark 47.* Two further points. First, a protocol that optimises independently over  $\mathbb{U}(6) \times \mathbb{U}(6)$  on each pair of blocks cannot test unbiasedness at all: mutual unbiasedness of a pair of bases is invariant under a *common* unitary, whereas independent local unitaries carry any pair of bases to any other. Second, the block rank profiles distinguish the two representatives:  $\mathcal{U}_1$  has 30 blocks of rank 6 and 6 of rank 5, while  $\mathcal{U}_2$  has 18 of each. Rank profiles are invariant under relabelling of the index sets, so  $\mathcal{U}_1$  and  $\mathcal{U}_2$  are structurally distinct points, not two labellings of one solution — consistent with [6].

Observation 32(d) was mild evidence against a clean outcome here, and Proposition 46 confirms it: an object that refuses to factor along  $6 = 2 \cdot 3$  does not hand over a  $\mathbb{Z}/6$ -structured set of bases. The MUB(6) problem is untouched by these AME matrices.

**On the golden ratio.** The existence of the Butson solutions shows that  $\varphi$  is not forced by AME(4, 6): there are representatives whose entire coefficient field is  $\mathbb{Q}(\zeta_6) = \mathbb{Q}(\sqrt{-3})$ , arithmetically disjoint from the golden field  $\mathbb{Q}(\zeta_{20})$  since  $\mathbb{Q}(\zeta_{20}) \cap \mathbb{Q}(\zeta_6) = \mathbb{Q}$ . Proposition 18 removes the most natural mechanism by which  $\varphi$  might have entered globally (icosahedral character values via  $\mathbb{P}^1(\mathbb{F}_5)$ ), and Proposition 21 identifies where it does enter: the block-level relations  $a^2 + b^2 = c^2$ ,  $b/a = \varphi$ , whose solution is the icosahedral vertex direction  $(1, \varphi)$ . The golden ratio in [4] is thus a local datum of a particular frame, plausibly inherited from the seeded numerical continuation, rather than an invariant of the AME(4, 6) moduli space. A sharper form of the remaining question: is there a functional on that moduli space whose critical points are the golden and Butson representatives? Section 5.2 shows that the obvious candidate, support density, must first be replaced by its local-unitary orbit minimum  $F_1^*$ , for which Proposition 30 gives  $\frac{1}{36} < F_1^* \leq \frac{7}{81}$ .

**The arithmetic thread.** Section 6 is deliberately quarantined. Weight 36 and dimension 6 share the modulus and the field  $\mathbb{Q}(\sqrt{-3})$ , and the numeral 36 appears in both as  $|\Phi^+(E_6)|$ , as  $6^2$ , and as a weight; but we have no map between them and do not assert one. What the section does contribute is independent: a proof of the degree law  $\deg P_{12m} = m(m-1)$  replacing an empirical observation, an independently computed Hecke cubic whose discriminant corroborates a separately obtained sextic field discriminant through a shared 12-digit prime, and reproducible unconditional numerics for  $L(s, \chi_{-3})$ . If a bridge exists, the most likely place to look is the appearance of  $\mathbb{Q}(\sqrt{-3})$

on both sides: as the coefficient field of the Butson representatives, and as the CM field of the Eisenstein arithmetic.

## A Status of claims

| Claim                                                                                | Status                                                   | Evidence                                                                       |
|--------------------------------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------|
| Lemma 6, Theorem 7 (parity obstruction)                                              | <b>proved</b>                                            | character-sum argument;<br>exhaustive check $2 \leq d \leq 12$                 |
| Proposition 9 ( $\mathbb{Z}[\omega]$ twist is a subfamily; 12 of 1296)               | <b>proved</b>                                            | trace expansion + Lemma 6                                                      |
| Remark 11 (obstruction is group-theoretic, not ring-theoretic)                       | <b>proved (elementary direction)</b>                     | Hall–Paige; exhaustive check on $\mathbb{Z}/6, S_3$                            |
| Proposition 13 (bilinear over $A \Leftrightarrow$ linear orthomorphism)              | <b>proved</b>                                            | character sums; Schur complement                                               |
| Proposition 14 (bilinear AME(4, 4) over $(\mathbb{Z}/2)^2$ )                         | verified computationally                                 | explicit $16 \times 16$ ; six purities $= 1/16$                                |
| Corollary 15 (some abelian $A$ works iff $d \not\equiv 2 \pmod{4}$ )                 | <b>proved</b>                                            | Sylow 2 argument; checked to $ A  = 22$                                        |
| Remark 16 ( $\mathbb{Z}/4 \times \mathbb{Z}/2$ : complete mapping but no linear one) | <b>proved</b>                                            | $ \text{Aut}  = 8$ has no element of order 3                                   |
| Theorem 3 (a)–(d), $E_6$ MUB decomposition                                           | verified computationally; underlying structure classical | explicit construction from Cartan matrix; full $12 \times 12$ overlap table    |
| Proposition 31 ( $\mathcal{U}_1, \mathcal{U}_2$ are Butson AME(4, 6))                | verified (reproduces [7, 8])                             | rebuilt from $F_6$ and CNOT; purities and exact $\mu_6$ entries                |
| Proposition 18 ( $\text{Aut} = \mathbb{Z}/2$ , no monomial lift)                     | verified for this representative                         | exhaustive leg search; component-wise separability test                        |
| Proposition 21 (amplitude closed forms; $\neq$ Fibonacci data)                       | <b>proved (exact identities)</b>                         | closed forms checked symbolically and numerically                              |
| Proposition 23 (9 blocks: $8 + 1$ )                                                  | verified computationally                                 | support components; unitarity of each block                                    |
| Proposition 24 (row/col support asymmetry; column Schmidt rank 2)                    | verified computationally                                 | direct count; per-block reconciliation                                         |
| Published entry list is exactly 2-unitary as published                               | verified                                                 | $\ U^\dagger U - I\ _F = 3.4 \times 10^{-15}$ , likewise $U^R, U^\Gamma$       |
| Proposition 26 (local datum is $\sqrt{5}$ , $\varphi$ derived)                       | <b>proved</b>                                            | half-angle identities, $\cos 2\theta = -1/\sqrt{5}$                            |
| Corollary 27 (not Fibonacci $F$ -data)                                               | <b>proved</b>                                            | modulus multiset is a gauge invariant                                          |
| Exceptional block is <i>not</i> a Kronecker product                                  | <b>proved (exhaustive)</b>                               | all $4! \times 4!$ permutations searched                                       |
| Block is an $F$ -matrix of <i>some</i> fusion category                               | <b>open</b>                                              | constrained to the two row profiles                                            |
| Proposition 30 ( $F_1^* > 1/36$ )                                                    | <b>proved</b>                                            | monomial reduction + Tarry                                                     |
| $F_1^*$ (golden) $= 7/81$ , i.e. 112 is minimal                                      | <b>open</b>                                              | only $F_1^* \leq 7/81$ is established                                          |
| Observation 32 (a)–(d)                                                               | verified numerically                                     | rank/consistency over $\mathbb{F}_2, \mathbb{F}_3$ ; shift tests; Gram spectra |
| Proposition 34 ( $\deg P_{12m} = m(m-1)$ )                                           | <b>proved (generic)</b>                                  | valence formula; $m = 1$ consistency check                                     |

*continued on next page*



*Status of claims (continued)*

| Claim                                                                                      | Status                                             | Evidence                                                                |
|--------------------------------------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------|
| Proposition 35 (weight-36 Hecke cubic, $S_3$ , totally real)                               | <b>computed independently</b>                      | $T_2$ built from $q$ -expansions; disc factored                         |
| Sextic $\mathbb{Q}(j_0)$ : $\text{Gal} = 2 \wr S_3$ , $h_K = 16$ , $\text{Cl} \cong C_2^4$ | reported independently; <i>not</i> reproduced here | corroborated only via shared discriminant primes                        |
| Observation 38 (34 zeros; bias 1.0428 at $x = 269$ )                                       | verified numerically                               | Hardy $Z$ from Hurwitz zetas; direct $\psi$ sum                         |
| Observation 41 (zeros predict bias variance 0.1132)                                        | verified numerically                               | 78 zeros to $T = 150$ ; dyadic windows to $10^7$                        |
| Remark 42 (second moment is independent of LI)                                             | <b>proved</b>                                      | orthogonality of cos kills cross terms for $\gamma \neq \gamma'$        |
| Observation 43 (fourth moment 0.0364, kurtosis 2.839)                                      | verified numerically                               | agrees to 1.5%; mild positive LI evidence                               |
| Observation 44 (full distribution via $\prod J_0(a_\gamma t)$ )                            | verified numerically                               | KS distance 0.034; tails match over 2 decades                           |
| Observation 45 (GUE repulsion; mean gap 0.99988)                                           | verified numerically                               | 189 zeros to $T = 300$ ; Poisson excluded                               |
| Remark 37 (sextic symbol defined exactly off primes $\mid 6$ )                             | <b>proved</b>                                      | $N(\varpi) \equiv 1 \pmod 6$ fails only at norms 3, 4                   |
| GRH for $L(s, \chi_{-3})$ at conductor 3 to height $3.3 \times 10^7$                       | <b>proved, not by us</b>                           | Platt [17], interval arithmetic                                         |
| Golden $\varphi$ is an artifact of the seed                                                | <b>conjectural</b>                                 | supported by Prop. 18, 21 and existence of Butson solutions; not proved |
| A bridge between Sections 3–5 and 6                                                        | <b>not claimed</b>                                 | shared modulus only                                                     |
| Proposition 46 (no block of $\mathcal{U}_1, \mathcal{U}_2$ is a basis)                     | verified computationally                           | all 36 block spectra, both decompositions                               |
| $\mathcal{U}_1 \not\cong \mathcal{U}_2$ (block rank profiles 30/6 vs 18/18)                | verified                                           | rank profile is relabelling-invariant                                   |
| MUB(6) data inside $\mathcal{U}_1$                                                         | <b>settled negatively</b>                          | Prop. 46                                                                |
| Remark 5 ( $E_6$ ’s 36 cannot index an AME(4, 6))                                          | verified computationally                           | $12 \times 3$ vs $6 \times 6$ ; 216 vanishing $h$ ; two-graph spectrum  |

## B Reproducibility

All results were obtained with NumPy in exact-integer or double precision as appropriate. Root systems were generated by reflection closure from the Cartan matrix; AME matrices were rebuilt from published defining data ( $\Lambda_j$  vectors, or the analytic entry list) and re-verified rather than trusted. Tolerances:  $10^{-9}$  for unitarity and purity assertions; observed errors were at the  $10^{-15}$  level. Exhaustive searches (Table 1, Proposition 18) enumerate their full stated domains.

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